

MATHEMATICS AND STATISTICS

TIME: 3 HOURS

[MAX. MARKS: 80]

Note: (i) All questions are compulsory.

(ii) Figures to the right indicate full marks.

(iii) Graph of L.P.P should be drawn on graph paper only.

(iv) Use of logarithmic table is allowed.

(v) Answers to the questions of Section – I and Section –II should be written in the only one answer book.

(vi) Answer to every new question must be written on a new page.

SECTION -I

Question 1.

(A) Select and write the appropriate answer from the given alternatives in each of the following sub-equation:

(i) If the points A(2,1,1), B(0,-1,4) and C(K,3,-2) are collinear, then k =

(a) 0

(b) 1

(c) 4

(d) -4

Sol:

Given that,

a(2,1,1), b(0,-1,4) and c(K,3,-2)

Since these points are collinear

$$\Rightarrow AB = \lambda AC$$

$$b - a = \lambda(c - a)$$

$$-2\hat{i} - 2\hat{j} + 3\hat{k} = \lambda[(k-2)\hat{i} + 2\hat{j} - 3\hat{k}]$$

Now, the comparing both sides, we get

$$2\lambda = -2 \Rightarrow \lambda = -1$$

$$\lambda(k-2) = -2 \Rightarrow -1(k-2) = -2$$

$$k-2 = 2$$

$$k = 4$$

Hence, the correct option is (c).

(ii) The inverse of the matrix $\begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$ is

(a) $\frac{1}{13} \begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}$

(b) $\frac{1}{13} \begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$

(c) $\frac{1}{13} \begin{bmatrix} -1 & -3 \\ 5 & 2 \end{bmatrix}$

(d) $\frac{1}{13} \begin{bmatrix} 1 & 5 \\ 3 & -2 \end{bmatrix}$

Sol:

Let,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\text{so, } \begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}^{-1} = \frac{1}{-2 - (-15)} \begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}$$

$$= \frac{1}{13} \begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}$$

Hence, the correct option is (a).

(iii) In $\triangle ABC$ if $a=13, b=14$ and $c=15$ then $\sin\left(\frac{A}{2}\right) = \dots\dots\dots$

Sol:

Given that,

$$a=13, b=14, c=15$$

$$s = \frac{a+b+c}{2} = \frac{13+14+15}{2} = \frac{42}{2} = 21$$

$$\begin{aligned}\sin \frac{A}{2} &= \sqrt{\frac{(s-b)(s-c)}{bc}} \\ &= \sqrt{\frac{(21-14)(21-15)}{14 \times 15}} \\ &= \sqrt{\frac{7 \times 6}{14 \times 15}} \\ &= \sqrt{\frac{1}{5}}\end{aligned}$$

Hence, the correct answer from the given alternative is (b).

(B) Attempt any THREE of the following:

(i) Find the volume of the parallelepiped whose coterminous edges are given by vectors $\hat{i} + 3\hat{j} - 4\hat{k}$, $5\hat{i} + 7\hat{j} + 5\hat{k}$ and $4\hat{i} + 5\hat{j} - 2\hat{k}$.

Sol:

Given that,

$$\vec{a} = \hat{i} + 3\hat{j} - 4\hat{k}$$

$$\vec{b} = 5\hat{i} + 7\hat{j} + 5\hat{k}$$

$$\vec{c} = 4\hat{i} + 5\hat{j} - 2\hat{k}$$

be the coterminous edges of the parallelepiped

Volume of the parallelepiped

$$\begin{aligned}[\vec{a}\vec{b}\vec{c}] &= \begin{vmatrix} 2 & 3 & -4 \\ 5 & 7 & 5 \\ 4 & 5 & -2 \end{vmatrix} \\ &= 2(-14 - 25) - 3(-10 - 20) - 4(25 - 28) \\ &= 2(-39) - 3(-30) - 4(-3) \\ &= -78 + 90 + 12 \\ &= 24 \text{ cubic units.}\end{aligned}$$

(ii) In $\triangle ABC$, prove that,) prove that $a(b \cos C - \cos B) = b^2 - c^2$.

Sol:

$$\begin{aligned}
 LHS &= a(b \cos C - \cos B) \\
 &= a \left[b \left(\frac{a^2 + b^2 - c^2}{2ab} \right) - c \cdot \left(\frac{a^2 + c^2 - b^2}{2ac} \right) \right] \\
 &= \left(\frac{a^2 + b^2 - c^2 - a^2 - c^2 + b^2}{2a} \right) \\
 &= \frac{2b^2 - 2c^2}{2} = \frac{2(b^2 - c^2)}{2} = b^2 - c^2
 \end{aligned}$$

Hence, the LHS = RHS.

(iii) If from a point Q(a,b,c) perpendiculars QA and QB are drawn to the YZ and ZX planes respectively, then find the vector equation of the plane OAB.

Sol:

Let the coordinates of the points A and B are (0, b, c) and (a, 0, c) respectively

$$\overrightarrow{OA} = b\hat{j} + c\hat{k}$$

$$\text{and, } \overrightarrow{OB} = a\hat{i} + c\hat{k}$$

The plane OAB passes through the point o($\vec{0}$) and is perpendicular to $\vec{n} = \overrightarrow{OA} \times \overrightarrow{OB}$

$$\begin{aligned}
 \vec{n} = \overrightarrow{OA} \times \overrightarrow{OB} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & b & c \\ a & 0 & c \end{vmatrix} \\
 &= bc\hat{i} + ac\hat{j} - ab\hat{k}
 \end{aligned}$$

Therefore, the equation of the plane OAB is,

$$(\vec{r} - \vec{0}) \cdot \vec{n} = 0$$

$$\vec{r} \cdot \vec{n} = 0$$

$$\vec{r}(bc\hat{i} + ac\hat{j} - ab\hat{k}) = 0$$

(iv) Find the Cartesian equation of the line passing through the points A(3, 4, -7) and B(6, -1, 1).

Sol:

The equation of the line passing through $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ is

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$(x_1, y_1, z_1) \equiv (3, 4, -7)$$

and,

$$(x_2, y_2, z_2) \equiv (6, -1, 1)$$

So, the required equation is

$$\frac{x-3}{6-3} = \frac{y-4}{-1-4} = \frac{z+7}{1+7}$$

$$\therefore \frac{x-3}{3} = \frac{y-4}{-5} = \frac{z+7}{8}$$

(v) Write the following statement in symbolic form and find its truth value:

$\forall n \in N, n^2 + n$ is an even number and $n^2 + n$ is an even number and $n^2 - n$ is an odd number.

Sol:

Let $P : n^2 + n$ is an even no. $\forall n \in N$

$q : n^2 - n$ is an odd no. $\forall n \in N$

So, the required symbolic form is $p \wedge q$

Truth value of p is True (T) and of q is False (F).

Therefore, truth value of $p \wedge q = F$

Question 2.

(A) Attempt any TWO Of the following:

(i) Using truth tables, examine whether the statement pattern $(p \wedge q) \vee (p \wedge r)$ is a tautology, contradiction or contingency.

Sol:

P	q	r	$p \wedge q$	$p \wedge r$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T
T	T	F	T	F	T
T	F	T	F	T	T
T	F	F	F	F	F
F	T	T	F	F	F
F	T	F	F	F	F
F	F	T	F	F	F
F	F	F	F	F	F

The last column of the above tables contains some entries of T and some of F.

Hence, the required statement is a contingency.

(ii) Find the shortest distance between the lines:

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \text{ and } \frac{x-}{2} = \frac{y-4}{4} = \frac{z-5}{5}$$

Sol:

Given that equation of the line are,

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \text{ and } \frac{x-}{2} = \frac{y-4}{4} = \frac{z-5}{5}$$

Here,

$$x_1 = 1, y_1 = 2, z_1 = 3$$

$$x_2 = 2, y_2 = 4, z_2 = 5$$

$$a_1 = 2, b_1 = 3, c_1 = 4$$

$$a_2 = 3, b_2 = 4, c_2 = 5$$

Now, distance between the lines is,

$$\begin{aligned} & \frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1 c_2 - c_1 b_2)^2 + (c_1 a_2 - c_2 a_1)^2 + (a_1 b_2 - a_2 b_1)^2}} \\ &= \frac{\begin{vmatrix} 1 & 2 & 2 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}}{\sqrt{(15-16)^2 + (12-10)^2 + (8-9)^2}} \\ &= \frac{1(15-16) - 2(10-12) + 2(8-9)}{\sqrt{1+4+1}} \\ &= \frac{-1+4-2}{\sqrt{6}} = \frac{1}{\sqrt{6}} \text{ units.} \end{aligned}$$

(iii) Find the general solution of the equation

$$\sin 2x + \sin 4x + \sin 6x = 0$$

Sol:

We have,

$$\sin 2x + \sin 6x + \sin 4x = 0$$

$$\Rightarrow (\sin 2x + \sin 6x) + \sin 4x = 0$$

$$\Rightarrow 2 \sin \frac{2x+6x}{2} \cdot \cos \frac{2x-6x}{2} + \sin 4x = 0$$

$$\Rightarrow 2 \sin 4x \cdot \cos(-2x) + \sin 4x = 0$$

$$\Rightarrow 2 \sin 4x \cdot \cos 2x + \sin 4x = 0$$

$$\Rightarrow \sin 4x(2 \cos 2x + 1) = 0$$

$$\Rightarrow \sin 4x = 0 \text{ or } 2 \cos 2x + 1 = 0$$

$$\Rightarrow \sin 4x = 0 \text{ or } \cos 2x = -\frac{1}{2}$$

$$\Rightarrow \sin 4x = 0 \text{ or } \cos 2x = -\cos \frac{\pi}{3}$$

$$\Rightarrow \sin 4x = 0 \text{ or } \cos 2x = \cos \frac{2\pi}{3}$$

$$\Rightarrow 4x = n\pi \text{ or } 2x = 2m\pi \pm \frac{2\pi}{3}$$

$$\Rightarrow x = \frac{n\pi}{4} \text{ or } x = m\pi \pm \frac{\pi}{3}$$

Are the general solutions where $n, m \in \mathbb{Z}$.

(B) Attempt any TWO Of the following:

(i) Solving the following equation by method of reduction:

$$x - y + z = 4, 2x + y - 3z = 0, x + y + z = 2$$

(ii) If θ is the measure of the acute angle between the lines represented by the equation

$$ax^2 + 2hxy + by^2 = 0, \text{ then prove that } \tan \theta = \left| \frac{2\sqrt{h^2 - ab}}{a+b} \right| \text{ where } a+b \neq 0 \text{ and } b \neq 0. \text{ Find the}$$

condition for coincident lines.

Sol:

Let m_1 and m_2 be the slope of the line represented by $ax^2 + 2hxy + by^2 = 0$

$$\therefore m_1 + m_2 = \frac{-2b}{b}, m_1 m_2 = \frac{a}{b} \text{ where } b \neq 0$$

Consider,

$$\begin{aligned}
 (m_1 - m_2)^2 &= (m_1 + m_2)^2 - 4m_1m_2 \\
 &= \left(\frac{-2h}{b}\right)^2 - 4 \times \frac{a}{b} \\
 &= \frac{4h^2}{b^2} - \frac{4a}{b} \\
 &= \frac{4h^2 - 4ab}{b^2} \\
 &= \frac{4}{b^2}(h^2 - ab) \\
 \therefore m_1 - m_2 &= \pm \frac{2}{b}\sqrt{h^2 - ab}
 \end{aligned}$$

Let θ be the acute angle between the lines.

$$\begin{aligned}
 \therefore \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 \cdot m_2} \right| \\
 &= \left| \frac{\pm \frac{2}{b}\sqrt{h^2 - ab}}{1 + \frac{a}{b}} \right| \\
 \therefore \tan \theta &= \left| \frac{2 \cdot \sqrt{h^2 - ab}}{a + b} \right| \\
 \therefore \theta &= \tan^{-1} \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right| \text{ where } a+b \neq 0
 \end{aligned}$$

If two lines are coincident, then $\theta = 0^\circ$, $\tan \theta = \tan 0^\circ = 0$

$$\begin{aligned}
 \therefore \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right| &= 0 \\
 \therefore h^2 - ab &= 0
 \end{aligned}$$

$\therefore$ The condition for coincident lines is $h^2 - ab = 0$.

(iii) Using vector method, find the indenture of the triangle whose vertices are P(0, 4, 0), Q(0, 0, 3) and R(0, 4, 3).

Question 3

(A) Attempt any TWO Of the following:

(i) Construct the switching circuit for the statement

Sol:

Let P: switch S_1 is closed.

q: switch S_2 is closed.

Therefore, the required switching circuit is,

(figher)

(ii) Find the joint equation of the pair of lines passing through the origin which are perpendicular respectively to the lines represented by $5x^2 + 2xy - 3y^2 = 0$.

Sol:

Let us consider the given equation,

$$5x^2 + 2xy - 3y^2 = 0$$

Comparing with $ax^2 + 2hxy - by^2 = 0$, we get

$$a = 5, b = -3, 2h = 2$$

Let m_1 and m_2 be the slope of the lines represented by the given equation.

$$m_1 + m_2 = \frac{-2h}{b} = \frac{-2}{-3} = \frac{2}{3}$$

$$m_1 \cdot m_2 = \frac{a}{b} = \frac{-5}{-3} = \frac{5}{3}$$

Since the required lines are perpendicular to these lines, their slopes are $-\frac{1}{m_1}$ and $-\frac{1}{m_2}$

Also, the required lines pass through the origin. So the equation are

$$y = -\frac{1}{m_1}x \text{ and } y = -\frac{1}{m_2}x$$

$$x + m_1y = 0 \text{ and } x + m_2y = 0$$

Therefore the equation is

$$(x + m_1 y)(x + m_2 y) = 0$$

$$\Rightarrow x + (m_1 + m_2)xy + m_1 m_2 y^2 = 0$$

$$\Rightarrow x^2 + \frac{2}{3}xy - \frac{5}{3}y^2 = 0$$

$$3x^2 + 2xy - 5y^2 = 0$$

Hence, the required equation is $3x^2 + 2xy - 5y^2 = 0$.

(iii) Show that $\cos^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) = \cos^{-1}\left(\frac{33}{65}\right)$

Sol:

Let $\cos^{-1}\left(\frac{4}{5}\right) = x$ and $\cos^{-1}\left(\frac{12}{13}\right) = y$ where $0 < x, y \leq \frac{\pi}{2}$

$$\cos x = \frac{4}{5} \text{ and } \cos y = \frac{12}{13}$$

$$\sin^2 x = 1 - \cos^2 x$$

$$1 - \frac{16}{25} = \frac{9}{25}$$

$$\therefore \sin x = \frac{3}{5}$$

$$\sin^2 y = 1 - \cos^2 y$$

$$1 - \frac{144}{169} = \frac{25}{169}$$

$$\therefore \sin y = \frac{5}{13}$$

$$\therefore \cos(x + y) = \cos x \cdot \cos y - \sin x \cdot \sin y$$

$$= \frac{4}{5} \times \frac{12}{13} - \frac{3}{5} \times \frac{5}{13}$$

$$= \frac{48}{65} - \frac{15}{65} = \frac{33}{65} \text{ and } 0 < x + y < \pi$$

$$\therefore x + y = \cos^{-1}\left(\frac{33}{65}\right)$$

$$\therefore \cos^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) = \cos^{-1}\left(\frac{33}{65}\right)$$

Hence, the LHS = RHS proved.

(B) Attempt any TWO Of the following:

(i) If l, m, n are the direction cosines of a line, then prove that $l^2 + m^2 + n^2 = 1$. Hence find the direction angles of 135° and 45° with Y and Z axes respectively.

Sol:

Let l, m, n be the direction cosines of line L.

Let $P(x, y, z)$ be any point on line L such that $l(OP) = r$

$$\therefore \vec{r} = \overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$$

If α, β, γ be the direction angles of the line OP, then

$$l = \cos \alpha, m = \cos \beta, n = \cos \gamma$$

Now,

$$\overrightarrow{OP} = (x\hat{i} + y\hat{j} + z\hat{k}) \cdot \hat{i} = x$$

Also,

$$\overrightarrow{OP} \cdot \hat{i} = |\overrightarrow{OP}| \cdot \cos \alpha = r \cdot \cos \alpha$$

So,

$$x^2 + y^2 + z^2 = r^2 (\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma)$$

$$r^2 = r^2 (l^2 + m^2 + n^2)$$

$$l^2 + m^2 + n^2 = 1$$

Given that,

$$\beta = 135^\circ, \gamma = 45^\circ, \alpha = ?$$

We have,

$$\begin{aligned}\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma &= 1 \\ \Rightarrow \cos^2 \alpha + \cos^2(135^\circ) + \cos^2(45^\circ) &= 1 \\ \Rightarrow \cos^2 \alpha + \left(\frac{-1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 &= 1 \\ \Rightarrow \cos^2 \alpha + \frac{1}{2} + \frac{1}{2} &= 1 \\ \Rightarrow \cos^2 \alpha &= 0 \\ \Rightarrow \cos \alpha &= 0 \\ \Rightarrow \alpha &= 90\end{aligned}$$

Hence, the direction angle of line with X-axis is 90° .

(ii) Find the vector and Cartesian equations of the plane passing through the points A(1, 1, -2), B(1, 2, 1) and C(2, -1, 1).

Sol:

Let $\vec{a}, \vec{b}, \vec{c}$ be the p.v of the points A(1, 1, -2), B(1, 2, 1), C(2, -1, 1) respectively.

$$\begin{aligned}\therefore \vec{a} &= \hat{i} + \hat{j} - 2\hat{k} \\ \vec{b} &= \hat{i} + 2\hat{j} + \hat{k} \\ \vec{c} &= 2\hat{i} - \hat{j} + \hat{k} \\ \therefore \overline{AB} &= \vec{b} - \vec{a} = (\hat{i} + 2\hat{j} + \hat{k}) - (\hat{i} + \hat{j} - 2\hat{k}) \\ &= \hat{j} + 3\hat{k} \\ \overline{AC} &= \vec{c} - \vec{a} = (2\hat{i} - \hat{j} + \hat{k}) - (\hat{i} + \hat{j} - 2\hat{k}) \\ &= \hat{i} - 2\hat{j} + 3\hat{k} \\ \overline{AB} \times \overline{AC} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 3 \\ 1 & -2 & 3 \end{vmatrix} \\ &= \hat{i}(3+6) - \hat{j}(-3) + \hat{k}(-1)\end{aligned}$$

The vector equation of the plane passing through the given points

$$\begin{aligned}\vec{r}(\overline{AB} \times \overline{AC}) &= \vec{a}(\overline{AB} \times \overline{AC}) \\ \Rightarrow \vec{r}(9\hat{i} + 3\hat{j} - \hat{k}) &= (\hat{i} + \hat{j} - 2\hat{k})(9\hat{i} + 3\hat{j} - \hat{k}) \\ \Rightarrow \vec{r}(9\hat{i} + 3\hat{j} - \hat{k}) &= 9 + 3 + 2 \\ \vec{r}(9\hat{i} + 3\hat{j} - \hat{k}) &= 14\end{aligned}$$

For Cartesian from,

Putting $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ in (i), we get

$$(x\hat{i} + y\hat{j} - z\hat{k})(9\hat{i} + 3\hat{j} - \hat{k}) = 14$$

$$\Rightarrow 9x + 3y - z = 14$$

Required Cartesian form of the equation of the plane is $9x + 3y - z = 14$.

(iii) Solve the following L.P.P By graphical method:

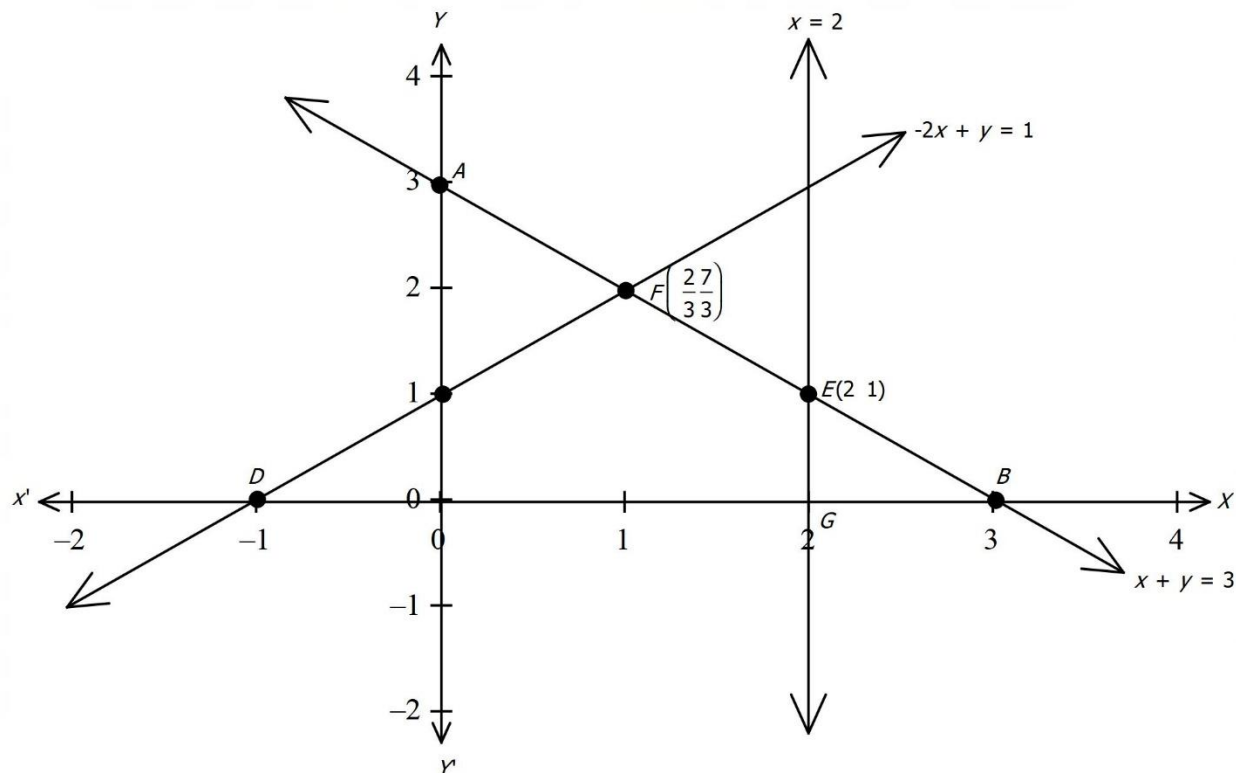
Maximize: $Z = 6x + 4y$

Subject to $x \leq 2, x + y \leq 3, -2x + y \leq 1, x \geq 0, y \geq 0$.

Sol:

We have,

Inequations	Equation	Points	Region
$x \leq 2$	$x = 2$	Line II to y axis passing through $x = 2$	Origin side
$x + y \leq 3$	$x + y = 3$	$A(0, 3), B(3, 0)$	Origin side
$-2x + y \leq 1$	$-2x + y = 1$	$C(0, 1), D\left(-\frac{1}{2}, 0\right)$	Origin side
$x \geq 0, y \geq 0$			1 st quadrant



Let E be the point of intersection of lines $x = 2$ and $x + y = 3$. Solving these equation we get,
 $E \equiv (2, 1)$

Let F be the point of intersection of lines $x + y = 3$ and $-2x + y = 1$

Solving these equations.

$$\begin{array}{r} x + y = 3 \\ -2x + y = 1 \\ + \quad - \quad - \end{array}$$

$3x = 2$

$$x = \frac{2}{3}$$

$$\begin{aligned}
 y &= 1 + 2x \\
 &= 1 + 2 \times \frac{2}{3} \\
 &= 1 + \frac{4}{3} \\
 &= \frac{7}{3} \\
 \therefore F &= \left(\frac{2}{3}, \frac{7}{3} \right), C(0,1)
 \end{aligned}$$

Therefore, Feasible region is shaded bounded region OGEFCO

The co-ordinates of the corner points are $O(0,0), G(2,0), E(2,1), F\left(\frac{2}{3}, \frac{7}{3}\right), C(0,1)$.

Now,

$$\begin{aligned}
 Z &= 6x + 4y \\
 Z(O) &= 0 + 0 = 0 \\
 Z(G) &= 6 \times 2 + 0 = 12 \\
 Z(E) &= 6 \times 2 + 4 \times 1 = 12 + 4 = 16 \\
 Z(F) &= 6 \times \frac{2}{3} + 4 \times \frac{7}{3} = 4 + \frac{28}{3} = \frac{40}{3} = 13.33 \\
 Z(C) &= 6 \times 0 + 4 \times 1 = 0 + 4 = 4
 \end{aligned}$$

Then,

Max. Value of Z is 16 at $E(2,1)$

Optimum solution is $x = 2$ and $y = 1$ and max. value of $Z = 16$

SECTION II

Question 4.

(A) Select and write the appropriate answer from the given alternatives in each of the following sub-questions:

(i) Derivative of $\tan^3 \theta$ with respect to $\sec^3 \theta$ at $\theta = \frac{\pi}{3}$ is.....

(a) $\frac{3}{2}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\frac{1}{2}$

(d) $\frac{-\sqrt{3}}{2}$

Sol:

Given that,

$$y = \tan^2 \theta, x = \sec^3 \theta$$

Differentiate both x and y w.r.to θ

$$\frac{dy}{d\theta} = 3 \tan^2 \theta \sec^2 \theta$$

$$\frac{dx}{d\theta} = 3 \sec^2 \theta \times \sec \theta \tan \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= \frac{3 \tan^2 \theta \times \sec^2 \theta}{3 \sec^2 \theta \times \sec \theta \tan \theta}$$

$$= \frac{\tan \theta}{\sec \theta} = \frac{\sin \theta}{\cos \theta} \times \cos \theta$$

$$= \sin \theta$$

$$\text{At } \theta = \frac{\pi}{3}$$

$$\frac{dy}{dx} = \sin \left(\frac{\pi}{3} \right) = \frac{\sqrt{3}}{2}$$

Hence, the correct option is (b).

(ii) The equation of tangent to the curve $y = 3x^2 - x + 1$ at $P(1, 3)$ is.....

(a) $5x - y = 2$

(b) $x + 5y = 16$

(c) $5x - y + 2 = 0$

(d) $5x = y$

Sol:

Let us consider the given curve is,

$$y = 3x^2 - x + 1$$

$$\frac{dy}{dx} = 6x - 1$$

At $P(1, 3)$

$$\frac{dy}{dx} = 5$$

Equation of tangent is at P(1,3) is

$$y - 3 = \frac{dy}{dx}(x - 1)$$

$$\Rightarrow y - 3 = 5(x - 1)$$

$$\Rightarrow y - 3 = 5x - 5$$

$$\Rightarrow 5x - y = 2$$

Hence, the correct option is (a)

(iii) The expected value of the number of heads obtained when three fair coins are tossed simultaneously is.....

(a) 1

(b) 1.5

(c) 0

(d) -1

Sol:

No. of heads x_i	Probability (p_i)	Sample space
0	$\frac{1}{8}$	TTT
1	$\frac{3}{8}$	TTH, HTT, THT
2	$\frac{3}{8}$	HHT, HTH, THH
3	$\frac{1}{8}$	HHH

$$\begin{aligned} \sum_{i=1}^n x_i p_i \\ \text{Expected value} &= 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8} \\ E(x) &= \frac{3}{8} + \frac{6}{8} + \frac{3}{8} = \frac{12}{8} \\ &= \frac{6}{4} = 1.5 \end{aligned}$$

(B) Attempt any THREE of following:

(i) Find $\frac{dy}{dx}$ if $x \sin y + y \sin x = 0$

Sol:

Given that,

$$x \sin y + y \sin x = 0$$

Differentiating w.r.to 'x',

$$x \cos y \frac{dy}{dx} + \sin y + y \cdot \cos x + \sin x \frac{dy}{dx} = 0$$

$$\therefore (x \cos y + \sin x) \frac{dy}{dx} = -(\sin y + y \cos x)$$

$$\therefore \frac{dy}{dx} = \frac{-(y \cos x + \sin y)}{(x \cos y + \sin x)}$$

(ii) Test whether the function $f(x) = x - \frac{1}{x}, x \in R, x \neq 0$,is increasing or decreasing.

Sol:

Given that,

$$f(x) = x - \frac{1}{x}$$

$$f'(x) = 1 + \frac{1}{x^2}$$

Now $x^2 \geq 0$ for all $x \in R$.

$$\therefore 1 + \frac{1}{x^2} > 0, \text{ for all } x \in R$$

$$\therefore f'(x) > 0$$

Hence, $f(x)$ is increasing.

(iii) Evaluate $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$

Sol:

Let,

$$I = \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$$

Now putting $\sqrt{x} = t$

$$\begin{aligned}\therefore \frac{1}{2\sqrt{x}} dx &= dt \\ \Rightarrow \frac{1}{\sqrt{x}} \cdot dx &= 2 \cdot dt \\ \therefore I &= \int \sin t \cdot 2 dt \\ &= 2 \int \sin t dt \\ &= 2(-\cos t) + c \\ &= -2 \cos \sqrt{x} + c\end{aligned}$$

(iv) Form the differential equation by eliminating arbitrary constants from the relation
 $y = Ae^{5x} + Be^{-5x}$.

Sol:

Given that,

$$\begin{aligned}\therefore \frac{dy}{dx} &= 5 \cdot A \cdot e^{5x} + 5B \cdot e^{-5x} \\ \frac{d^2y}{dx^2} &= 25Ae^{5x} + 25Be^{-5x} \\ &= 25(Ae^{5x} + Be^{-5x}) \\ &= \frac{d^2y}{dx^2} = 25y\end{aligned}$$

Hence, the required differential equation is

$$\frac{d^2y}{dx^2} - 25y = 0$$

(v) The probability that a bomb will hit a target is 0.8. Find the probability that out of 10 bombs dropped, exactly 4 will hit the target.

Sol:

Given that,

x = No. of bombs hitting the target

p = Probability of hitting the target

Here $p = 0.8, q = 1 - p = 1 - 0.8 = 0.2, n = 10, x = 4$

$$\therefore X \sim B(n, p)$$

$$\therefore X \sim B(10, 0.8)$$

$$P(X = x) = {}^nC_x \cdot p^x \cdot q^{n-x}$$

$$\therefore P(X = 4) = {}^{10}C_4 \cdot (0.8)^4 (0.2)^{10-4}$$

$$= {}^{10}C_4 \cdot (0.8)^4 (0.2)^6$$

$$= \frac{10!}{4!(10-4)!} \times 0.4096 \times 0.000064$$

$$= 210 \times 0.4096 \times 0.000064$$

$$= 0.005505$$

Question 5

(A) Attempt any TWO of following:

(i) Solve $\frac{dy}{dx} = \cos(x + y)$.

Sol:

We have,

$$\frac{dy}{dx} = \cos(x + y)$$

Let,

$$x + y = u$$

Differentiating w.r.to 'x'.

$$1 + \frac{dy}{dx} = \frac{du}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{du}{dx} - 1$$

$$\Rightarrow \cos u = \frac{du}{dx} - 1$$

$$\Rightarrow \frac{du}{dx} = \cos u + 1$$

$$\Rightarrow \frac{du}{dx} = 2 \cos^2 \frac{u}{2}$$

$$\Rightarrow \frac{1}{\cos^2 \frac{u}{2}} \cdot du = 2 dx$$

$$\Rightarrow \sec^2 \frac{u}{2} \cdot du = 2 dx$$

$$\Rightarrow \int \sec^2 \frac{u}{2} \cdot du = 2 \int dx$$

$$\Rightarrow 2 \tan \frac{u}{2} = 2x + c'$$

$$\Rightarrow \tan \frac{u}{2} = x + \frac{c'}{2}$$

$$\Rightarrow \tan \left(\frac{x+y}{2} \right) = x + c \text{ where } c = \frac{c'}{2}$$

(ii) If u and v are two functions of x, then prove that:

$$\int u v dx = u \int v dx - \int \left[\frac{du}{dx} \int v dx \right] dx$$

Sol:

Let,

$$\int v \cdot dx = w$$

$$\therefore \frac{dw}{dx} = v$$

Now,

$$\begin{aligned}\frac{d}{dx}(u \cdot w) &= u \cdot \frac{dw}{dx} + w \cdot \frac{du}{dx} \\ &= u \cdot v + \int v \cdot dx \cdot \frac{du}{dx} \\ &= uv + \frac{du}{dx} \int v \cdot dx\end{aligned}$$

Now, integrating both sides w.r.t 'x', we get

$$\begin{aligned}u \cdot w &= \int u \cdot v dx + \int \left[\frac{du}{dx} \int v \cdot dx \right] \cdot dx \\ \therefore u \cdot \int v \cdot dx &= \int u \cdot v dx + \int \left[\frac{du}{dx} \int v \cdot dx \right] \cdot dx \\ \int u \cdot v dx &= u \int v \cdot dx - \int \left[\frac{du}{dx} \int v \cdot dx \right] dx\end{aligned}$$

(iii) If $f(x) = \frac{e^{x^2} - \cos x}{x^2}$, for $x \neq 0$ is continuous at $\frac{d^2y}{dx^2} - 25y = 0$ find $f(0)$.

Sol:

Since $f(x)$ is continuous at $x = 0$, we get

$$\begin{aligned}
 f(0) &= \lim_{x \rightarrow 0} f(x) \\
 f(0) &= \lim_{x \rightarrow 0} \frac{e^{x^2}}{x^2} \\
 &= \lim_{x \rightarrow 0} \frac{(e^{x^2} - \cos x)}{x^2} \\
 &= \lim_{x \rightarrow 0} \frac{(e^{x^2} - 1) + 2 \sin^2 \frac{x}{2}}{x^2} \\
 &= \lim_{x \rightarrow 0} \frac{(e^{x^2} - 1)}{x^2} + \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{x^2} \\
 &= 1 + 2 \cdot \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{2}}{\left(\frac{x}{2}\right)^2 \times \frac{1}{4}} \\
 &= 1 + 2 \cdot \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{2}}{\frac{1}{2}} \right)^2 \times \frac{1}{4} \\
 &= 1 + 2 \times 1 \times \frac{1}{4} \\
 &= 1 + \frac{1}{2} = \frac{3}{2} \\
 \therefore f(0) &= \frac{3}{2}
 \end{aligned}$$

(B) Attempt any THREE of following:

(i) If $y = f(x)$ is a differentiable function of x such that inverse function $x = f^{-1}(y)$ exists, then prove that x is a differentiable function of y and $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$ where $\frac{dy}{dx} \neq 0$ hence find

$$\frac{d}{dx}(\tan^{-1} x).$$

Sol:

Let δx be an increment in x .

δy be the corresponding increment in y .

$$\therefore \frac{\delta y}{\delta x} \times \frac{\delta x}{\delta y} = 1$$

$$\therefore \frac{\delta x}{\delta y} = \frac{1}{\left(\frac{\delta y}{\delta x}\right)}; \frac{\delta y}{\delta x} \neq 0$$

$$\lim_{\delta x \rightarrow 0} \frac{\delta x}{\delta y} = \lim_{\delta x \rightarrow 0} \frac{1}{\left(\frac{\delta y}{\delta x}\right)} = \frac{1}{\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x}}$$

As $\delta x \rightarrow 0 \Rightarrow \delta y \rightarrow 0$

$$\lim_{\delta x} \frac{\delta x}{\delta y} = \frac{1}{\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x}}$$

Y is differentiable function of x,

$$\frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x}$$

$$\therefore \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \frac{1}{\frac{dy}{dx}}; \frac{dy}{dx} \neq 0$$

So, the limit on R.H.S exists, limit on L.H.S also exists and is equal to $\frac{dx}{dy}$.

$$\frac{dy}{dx} = \frac{1}{\frac{dy}{dx}}; \frac{dy}{dx} \neq 0$$

Therefore, x is a differentiable function of y.

Let,

$$y = \tan^{-1} x$$

$$\therefore x = \tan y$$

Differentiating w.r.t. 'x',

$$\frac{dx}{dy} = \sec^2 y = 1 + \tan^2 y = 1 + x^2$$

now,

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{1+x^2}$$

$$\therefore \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

(ii) A telephone company in a town has 5000 subscribers on its list and collects fixed rent charges of ₹3,000 per year from each subscriber. The company proposes to increase annual rent and it is believed that for every increase of one rupee in the rent, one subscriber will be discontinued. Find what increased annual rent will bring the maximum annual income to the company.

Sol:

Let us consider the annual rental be increased by ₹ x.

X subscribers will discontinue.

Let I be the annual income of the company.

$$\therefore I = (3000 + x)(5000 - x)$$

$$\begin{aligned}\therefore I'(x) &= (3000 + x)(-1) + (5000 - x) \times 1 \\ &= 2000 - 2x\end{aligned}$$

$$\therefore I''(x) = -2$$

For maximum income,

$$I'(x) = 0$$

$$2000 - 2x = 0$$

$$\therefore x = 1000$$

Now,

$$I''(x = 1000) = -2 < 0$$

Now, Income I is maximum at $x = 1000$.

And, annual income should be increased by ₹ 1000 for maximum annual income.

(iii) Evaluate: $\int_{-a}^a \sqrt{\frac{a-x}{a+x}} dx.$

Sol:

Let,

$$\begin{aligned} I &= \int_{-a}^a \sqrt{\frac{a-x}{a+x}} dx \\ &= \int_{-a}^a \sqrt{\frac{a-x}{a+x} \times \frac{a-x}{a-x}} dx \\ &= \int_{-a}^a \frac{a-x}{\sqrt{a^2-x^2}} dx \\ I &= \int_{-a}^a \frac{a}{\sqrt{a^2-x^2}} dx - \int_{-a}^a \frac{x}{\sqrt{a^2-x^2}} dx \end{aligned}$$

By property of even and odd function,

$$\begin{aligned} \int_{-a}^a f(x) dx &= 2 \int_0^a f(x) dx \text{ if } f(x) \text{ is even} \\ &= 0 \text{ if } f(x) \text{ is odd.} \end{aligned}$$

Here $\frac{a}{\sqrt{a^2-x^2}}$ is an even function and $\frac{x}{\sqrt{a^2-x^2}}$ is an odd function.

$$\begin{aligned} I &= \int_0^a \frac{a}{\sqrt{a^2-x^2}} dx - 0 \\ &= 2a \int_0^a \frac{1}{\sqrt{a^2-x^2}} dx \\ &= 2a \left[\sin^{-1} \left(\frac{x}{a} \right) \right]_0^a \\ &= 2a (\sin^{-1} 1 - \sin^{-1} 0) \\ &= 2a \left(\frac{\pi}{2} - 0 \right) = 2a \cdot \frac{\pi}{2} = \pi a \\ \therefore \int_{-a}^a \sqrt{\frac{a-x}{a+x}} \cdot dx &= \pi a \end{aligned}$$

Question 6

(A) Attempt any TWO of following:

(i) Discuss the continuity of the following function, at $x = 0$

$$f(x) = \frac{x}{|x|}, \text{ for } x \neq 0$$

$$= 1, \text{ for } x = 0$$

Sol:

Given that,

$$f(x) = \frac{x}{|x|}; \text{ for } x \neq 0$$

$$= 1; \text{ for } x = 0$$

$$\therefore \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{x}{|x|} = \lim_{x \rightarrow 0^-} \frac{x}{-x} = -1 \dots \dots \dots (1)$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x}{|x|} = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1 \dots \dots \dots (2)$$

From (1) and (2), we get

$$\therefore \lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$$

$$\therefore \lim_{x \rightarrow 0} f(x) \text{ does not exist}$$

$$\therefore f(x) \text{ is discontinuous at } x = 0$$

(ii) If the population of a country doubles in 60 years, in how many years will it be triple under the assumption that the rate of increase is proportion to the number of inhabitants? [Given: $\log 2 = 0.6912$ and $\log 3 = 1.0986$.]

Sol:

Let P be the population of the country at time t.

$$\frac{dp}{dt} \propto P$$

$$\Rightarrow \frac{dp}{dt} = KP$$

$$\Rightarrow \frac{dp}{dt} = K \cdot dt$$

Integrating both sides,

$$\int \frac{1}{P} \cdot dP = K \int dt$$

$$\therefore \log P = Kt + c \dots \dots \dots (1)$$

$$\text{Let } P_0 = P_0 \text{ at } t = 0$$

Substituting in (1)

$$\log P_0 = K \cdot 0 + c$$

$$\Rightarrow c = \log P_0$$

$$\Rightarrow \log P = Kt + \log P_0$$

$$\Rightarrow \log P - \log P_0 = Kt$$

$$\Rightarrow \log \frac{P}{P_0} = Kt \text{ -----(2)}$$

When,

$$t = 60 \text{ yrs}, P = 2P_0$$

$$\therefore \log \frac{2P_0}{P_0} = 60K$$

$$K = \frac{1}{60} \times \log 2$$

$$K = \frac{1}{60} \times 0.6912$$

Therefore, equation (2) can be written as

$$\log \frac{P}{P_0} = \frac{0.6912}{60} \times t \text{ -----(3)}$$

Now, when

$$P = 3P_0, t = ?$$

$$\log \frac{3P_0}{P_0} = \frac{0.6912}{60} \times t$$

$$\log 3 \frac{60}{0.6912} = t$$

$$t = 1.0986 \times \frac{60}{0.6912}$$

$$t = 95.36 \text{ yrs.}$$

(iii) A fair coin is tossed 8 times. Find the probability that it shows heads:

(a) exactly 5 times

(b) at least once.

(B) Attempt any TWO of following:

(i) Evaluate $\int \frac{d\theta}{\sin \theta - \sin 2\theta}$

Sol:

Given that,

$$\begin{aligned} I &= \int \frac{d\theta}{\sin \theta - \sin 2\theta} \\ &= \int \frac{d\theta}{\sin \theta + 2 \sin \theta \cdot \cos \theta} \\ &= \int \frac{d\theta}{\sin \theta (1 + 2 \cos \theta)} \\ &= \int \frac{\sin \theta \cdot d\theta}{\sin^2 \theta \cdot (1 + 2 \cos \theta)} \\ &= \int \frac{\sin \theta \cdot d\theta}{(1 - \cos^2 \theta) \cdot (1 + 2 \cos \theta)} \\ &= \int \frac{\sin \theta \cdot d\theta}{(1 + \cos \theta)(1 - \cos \theta)(1 + 2 \cos \theta)} \end{aligned}$$

Let,

$$\cos \theta = t$$

$$\therefore -\sin \theta \cdot d\theta = dt$$

$$\therefore I = \int \frac{-dt}{(1+t)(1-t)(1+2t)}$$

Consider,

$$\begin{aligned} \frac{-1}{(1+t)(1-t)(1+2t)} &= \frac{A}{(1+t)} + \frac{B}{(1-t)} + \frac{C}{(1+2t)} \\ \therefore -1 &= A(1-t)(1+2t) + B(1+t)(1+2t) + C(1+t)(1-t) \end{aligned}$$

Putting $t = 1$,

$$\therefore -1 = B(2 \times 3)$$

$$\therefore B = \frac{-1}{6}$$

Putting $t = -1$,

$$\therefore -1 = A(2)(-1)$$

$$\therefore A = \frac{1}{2}$$

$$\text{Putting } t = -\frac{1}{2},$$

$$\therefore -1 = c \left(1 - \frac{1}{2}\right) \left(1 + \frac{1}{2}\right)$$

$$\therefore -1 = c \left(\frac{1}{2}\right) \left(\frac{3}{2}\right) = c \left(\frac{3}{4}\right)$$

$$\therefore c = \frac{-4}{3}$$

$$\therefore \frac{-1}{(1+t)(1-t)(1+2t)} = \frac{\frac{1}{2}}{1+t} + \frac{\frac{-1}{6}}{1-t} + \frac{\frac{-4}{3}}{1+2t}$$

$$\therefore I = \int \frac{-dt}{(1+t)(1-t)(1+2t)}$$

$$= \int \left[\frac{1}{2(1+t)} - \frac{1}{6(1-t)} - \frac{4}{3(1+2t)} \right] dt$$

$$= \frac{1}{2} \int \frac{1}{1+t} \cdot dt - \frac{1}{6} \int \frac{1}{1-t} dt - \frac{4}{3} \int \frac{1}{1+2t} dt$$

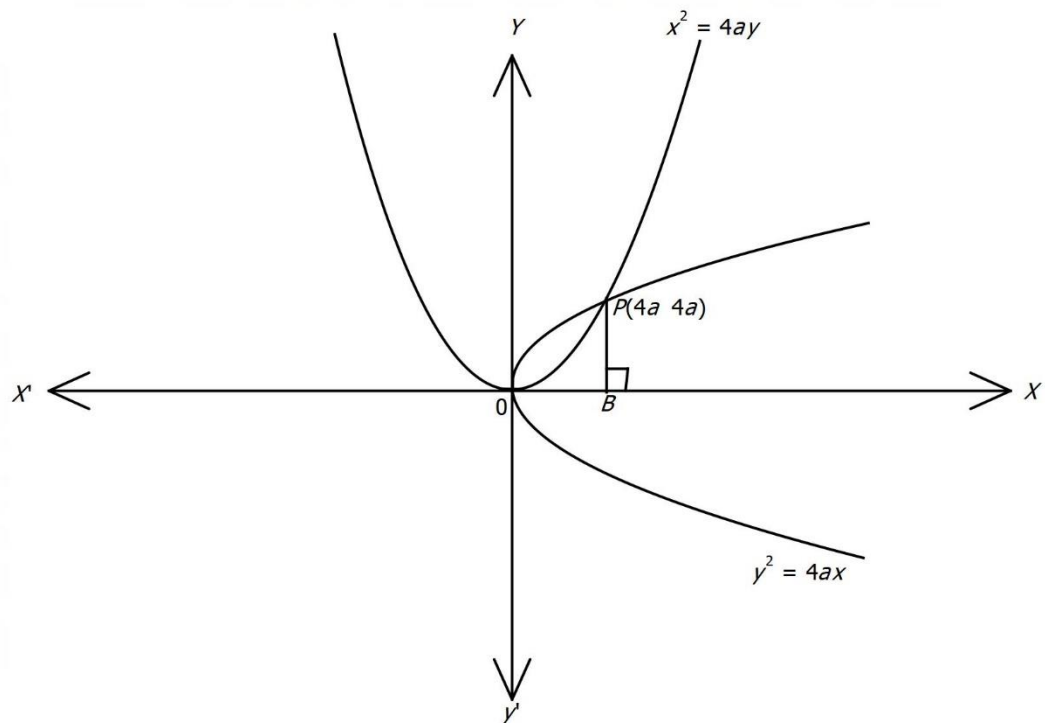
$$= \frac{1}{2} \log|(1+t)| + \frac{1}{6} \log|(1-t)| - \frac{4}{3 \times 2} \log|(1+2t)| + c$$

$$I = \frac{1}{2} \log|1 + \cos \theta| + \frac{1}{6} \log|1 - \cos \theta| - \frac{2}{3} \log|1 + 2 \cos \theta| + c$$

(ii) Find the area of the region lying between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$.

Sol:

Equation of the parabolas are $y^2 = 4ax$ and $x^2 = 4ay$



Solving the equations,

$$\left(\frac{x^2}{4a}\right)^2 = 4ax$$

$$\therefore x^4 = 64a^3x$$

$$\therefore x^4 - 64a^3x = 0$$

$$\therefore x(x^3 - 64a^3x) = 0$$

$$\therefore x = 0 \text{ or } x^3 = 64a^3$$

$$\therefore x^3 = (4a)^3$$

$$\therefore x = 4a$$

When,

$$x = 0$$

$$y^2 = 4ax = 0 \therefore y = 0$$

When,

$$x = 4a$$

$$y^2 = 4ax = 4a \times 4a \therefore y = 4a$$

$\therefore$ The points of intersection of the parabolas are $O(0, 0)$, $P(4a, 4a)$.

∴ The required area is bounded by $y^2 = 4ax$ and $x^2 = 4ay$ between $x=0$ and $x=4a$.

$$\begin{aligned}\therefore A &= \int_0^{4a} \sqrt{4ax} dx - \int_0^{4a} \frac{x^2}{4a} \cdot dx \\ &= \sqrt{4a} \int_0^{4a} \sqrt{x} dx - \frac{1}{4a} \int_0^{4a} x^2 \cdot dx \\ &= \frac{4\sqrt{a}}{3} [4a]^{\frac{3}{2}} - \frac{1}{12a} (4a)^3 \\ &= \frac{4\sqrt{a}}{3} \times 4a\sqrt{4a} - \frac{1}{12a} \times 64a^3 \\ &= \frac{32}{3} a^2 - \frac{16}{3} a^2 = \frac{16}{3} a^2 \text{ sq. unit}\end{aligned}$$

(iii) Given the probability density function (p.d.f) of a continuous random variable X as.

$$\begin{aligned}f(x) &= \frac{x^2}{3}, -1 < x < 2 \\ &= 0, \text{ otherwise.}\end{aligned}$$

Determine the cumulative distribution function (c.d.f) of X and hence find $P(X < 1)$, $P(X > 0)$, $P(1 < X < 2)$.

Sol:

The p.d.f of a continuous r.v.X is

$$\begin{aligned}f(x) &= \frac{x^2}{3}, -1 < x < 2 \\ &= 0, \text{ otherwise.}\end{aligned}$$

∴ c.d.f of X is given by

$$F(x) = \int_{-1}^x f(y) \cdot dy$$

$$= \int_{-1}^x \frac{y^2}{3} \cdot dy = \left[\frac{y^3}{9} \right]_{-1}^x$$

$$\therefore F(x) = \frac{x^3}{9} + \frac{1}{9} \forall x \in R.$$

$$\therefore P(X < 1) = F(1) = \frac{1}{9} + \frac{1}{9} = \frac{2}{9}$$

$$P(X > 0) = 1 - P(X \leq 0)$$

$$= 1 - F(0)$$

$$= 1 - \left(\frac{0}{9} + \frac{1}{9} \right)$$

$$= 1 - \frac{1}{9} = \frac{8}{9}$$

$$P(1 < K < 2) = F(2) - F(1)$$

$$= \frac{2^3}{9} + \frac{1}{9} - \left(\frac{1^3}{9} + \frac{1}{9} \right)$$

$$= \frac{8}{9} + \frac{1}{9} - \frac{1}{9} - \frac{1}{9}$$

$$= \frac{8}{9} - \frac{1}{9}$$

$$= \frac{7}{9}$$